Economic Growth, the Financial System, and Business Cycles

ECON 101 - Macroeconomics

Muhebullah Karimzada

Winter 2026

Chapter Outline

- **6.1** Long-Run Economic Growth
- **6.2** Saving, Investment, and the Financial System
- **6.3** The Business Cycle

Chapter Outline

- **6.1** Long-Run Economic Growth
- **6.2** Saving, Investment, and the Financial System
- **6.3** The Business Cycle

Chapter Outline

- **6.1** Long-Run Economic Growth
- **6.2** Saving, Investment, and the Financial System
- **6.3** The Business Cycle

6.1 Learning Objective

Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising productivity increases the **average standard of living**.
- This is different from short-run fluctuations (**business cycles**):
  - **Expansion:** real GDP rising.
  - **Recession:** real GDP falling.
- The most common measure of the standard of living is:
  - **Real GDP per capita** = real GDP divided by population.

6.1 Learning Objective

Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising productivity increases the **average standard of living**.
- This is different from short-run fluctuations (**business cycles**):
  - **Expansion:** real GDP rising.
  - **Recession:** real GDP falling.
- The most common measure of the standard of living is:
  - **Real GDP per capita** = real GDP divided by population.

6.1 Learning Objective

Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising productivity increases the **average standard of living**.
- This is different from short-run fluctuations (**business cycles**):
  - **Expansion:** real GDP rising.
  - **Recession:** real GDP falling.
- The most common measure of the standard of living is:
  - **Real GDP per capita** = real GDP divided by population.

Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising productivity increases the **average standard of living**.
- This is different from short-run fluctuations (**business cycles**):
  - **Expansion:** real GDP rising.
  - **Recession:** real GDP falling.
- The most common measure of the standard of living is:
  - **Real GDP per capita** = real GDP divided by population.

6.1 Learning Objective

Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising productivity increases the **average standard of living**.
- This is different from short-run fluctuations (**business cycles**):
  - **Expansion:** real GDP rising.
  - **Recession:** real GDP falling.
- The most common measure of the standard of living is:
  - **Real GDP per capita** = real GDP divided by population.

6.1 Learning Objective

Goal

Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising productivity increases the **average standard of living**.
- This is different from short-run fluctuations (**business cycles**):
    - **Expansion:** real GDP rising.
    - **Recession:** real GDP falling.
- The most common measure of the standard of living is:
    - **Real GDP per capita** = real GDP divided by population.

Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
  - How much goods and services, on average, each person can consume.
  - Adjusted for changes in the price level (**real**, not nominal).
- It does not capture everything about well-being, but:
  - Countries with higher real GDP per capita typically have better:
    - health, education, housing, and infrastructure.

Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
  - How much goods and services, on average, each person can consume.
  - Adjusted for changes in the price level (**real**, not nominal).
- It does not capture everything about well-being, but:
  - Countries with higher real GDP per capita typically have better:
    - health, education, housing, and infrastructure.

Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
  - How much goods and services, on average, each person can consume.
  - Adjusted for changes in the price level (**real**, not nominal).
- It does not capture everything about well-being, but:
  - Countries with higher real GDP per capita typically have better:
    - health, education, housing, and infrastructure.

Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
  - How much goods and services, on average, each person can consume.
  - Adjusted for changes in the price level (**real**, not nominal).
- It does not capture everything about well-being, but:
  - Countries with higher real GDP per capita typically have better:
    - health, education, housing, and infrastructure.

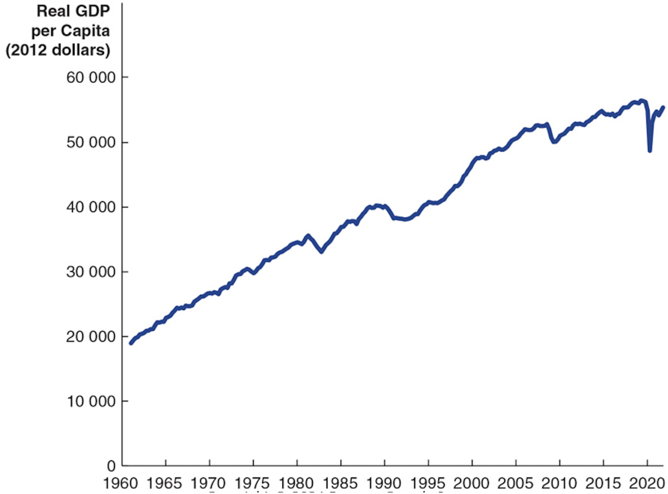
Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
  - How much goods and services, on average, each person can consume.
  - Adjusted for changes in the price level (**real**, not nominal).
- It does not capture everything about well-being, but:
  - Countries with higher real GDP per capita typically have better:
    - health, education, housing, and infrastructure.

Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
  - How much goods and services, on average, each person can consume.
  - Adjusted for changes in the price level (**real**, not nominal).
- It does not capture everything about well-being, but:
  - Countries with higher real GDP per capita typically have better:
    - health, education, housing, and infrastructure.

Figure 6.1 – Real GDP per Capita, 1961–2021



Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita in Canada has risen more than **three-fold** since 1961.
- Interpretation:
  - The average Canadian in 2021 can buy almost three times as many goods and services as the average Canadian in 1961.
- This long-run rise is what we mean by **economic growth**.
- Short-run recessions are bumps around this upward trend.

Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita in Canada has risen more than **three-fold** since 1961.
- Interpretation:
    - The average Canadian in 2021 can buy almost three times as many goods and services as the average Canadian in 1961.
- This long-run rise is what we mean by **economic growth**.
- Short-run recessions are bumps around this upward trend.

Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita in Canada has risen more than **three-fold** since 1961.
- Interpretation:
  - The average Canadian in 2021 can buy almost three times as many goods and services as the average Canadian in 1961.
- This long-run rise is what we mean by **economic growth**.
- Short-run recessions are bumps around this upward trend.

Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita in Canada has risen more than **three-fold** since 1961.
- Interpretation:
  - The average Canadian in 2021 can buy almost three times as many goods and services as the average Canadian in 1961.
- This long-run rise is what we mean by **economic growth**.
- Short-run recessions are bumps around this upward trend.

Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita in Canada has risen more than **three-fold** since 1961.
- Interpretation:
  - The average Canadian in 2021 can buy almost three times as many goods and services as the average Canadian in 1961.
- This long-run rise is what we mean by **economic growth**.
- Short-run recessions are bumps around this upward trend.

Calculating Growth Rates

- Growth rate of an economic variable (e.g., real GDP) from year $t-1$ to year t:

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, we can compute growth each year and then take the average.
- Example:
  - Real GDP per capita grows by 2%, 3%, and 1% over three years.
  - Average annual growth $\approx (2+3+1)/3 = 2\%$.

Calculating Growth Rates

- Growth rate of an economic variable (e.g., real GDP) from year $t-1$ to year t:

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, we can compute growth each year and then take the average.
- Example:
  - Real GDP per capita grows by 2%, 3%, and 1% over three years.
  - Average annual growth $\approx (2+3+1)/3 = 2\%$.

Calculating Growth Rates

- Growth rate of an economic variable (e.g., real GDP) from year $t-1$ to year t:

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, we can compute growth each year and then take the average.
- Example:
  - Real GDP per capita grows by 2%, 3%, and 1% over three years.
  - Average annual growth $\approx (2+3+1)/3 = 2\%$.

Calculating Growth Rates

- Growth rate of an economic variable (e.g., real GDP) from year $t-1$ to year t:

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, we can compute growth each year and then take the average.
- Example:
  - Real GDP per capita grows by 2%, 3%, and 1% over three years.
  - Average annual growth $\approx (2+3+1)/3 = 2\%$.

Calculating Growth Rates

- Growth rate of an economic variable (e.g., real GDP) from year $t-1$ to year t:

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, we can compute growth each year and then take the average.
- Example:
  - Real GDP per capita grows by 2%, 3%, and 1% over three years.
  - Average annual growth $\approx (2+3+1)/3 = 2\%$.

Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.
- Suppose:
  - Previous real GDP = Y_0,
  - Current real GDP = Y_t,
  - Annual growth rate = g,
  - Number of years between = t.
- Then:

$$Y_0(1+g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.
- Suppose:
  - Previous real GDP = Y_0,
  - Current real GDP = Y_t,
  - Annual growth rate = g,
  - Number of years between = t.
- Then:

$$Y_0(1+g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.
- Suppose:
  - Previous real GDP $= Y_0$,
  - Current real GDP $= Y_t$,
  - Annual growth rate $= g$,
  - Number of years between $= t$.
- Then:

$$Y_0(1+g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.
- Suppose:
  - Previous real GDP $= Y_0$,
  - Current real GDP $= Y_t$,
  - Annual growth rate $= g$,
  - Number of years between $= t$.
- Then:

$$Y_0(1+g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.
- Suppose:
  - Previous real GDP $= Y_0$,
  - Current real GDP $= Y_t$,
  - Annual growth rate $= g$,
  - Number of years between $= t$.
- Then:

$$Y_0(1+g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.
- Suppose:
  - Previous real GDP $= Y_0$,
  - Current real GDP $= Y_t$,
  - Annual growth rate $= g$,
  - Number of years between $= t$.
- Then:

$$Y_0(1+g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.
- Suppose:
  - Previous real GDP $= Y_0$,
  - Current real GDP $= Y_t$,
  - Annual growth rate $= g$,
  - Number of years between $= t$.
- Then:

$$Y_0(1+g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

Growth Rates over Longer Periods

- Over longer periods, we use **compound growth**.
- Suppose:
  - Previous real GDP $= Y_0$,
  - Current real GDP $= Y_t$,
  - Annual growth rate $= g$,
  - Number of years between $= t$.
- Then:

$$Y_0(1+g)^t = Y_t.$$

- Solving for g gives the average annual growth rate over the period.

The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years it takes to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- Example:
  - If real GDP per capita grows at 5% per year:
  - Years to double $\approx 70/5 = 14$ years.
- This shows how even small differences in growth rates matter a lot over time.

The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years it takes to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- Example:
  - If real GDP per capita grows at 5% per year:
  - Years to double $\approx 70/5 = 14$ years.
- This shows how even small differences in growth rates matter a lot over time.

The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years it takes to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- Example:
  - If real GDP per capita grows at 5% per year:
  - Years to double $\approx 70/5 = 14$ years.
- This shows how even small differences in growth rates matter a lot over time.

The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years it takes to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- Example:
  - If real GDP per capita grows at 5% per year:
  - Years to double $\approx 70/5 = 14$ years.
- This shows how even small differences in growth rates matter a lot over time.

The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years it takes to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- Example:
  - If real GDP per capita grows at 5% per year:
  - Years to double $\approx 70/5 = 14$ years.
- This shows how even small differences in growth rates matter a lot over time.

The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years it takes to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- Example:
  - If real GDP per capita grows at 5% per year:
  - Years to double $\approx 70/5 = 14$ years.
- This shows how even small differences in growth rates matter a lot over time.

Rule of 70: Classroom Example

- If Country A grows at 2% per year:
  - It takes about $70/2 = 35$ years to double income per person.
- If Country B grows at 4% per year:
  - It takes about $70/4 = 17.5$ years to double.
- After a few decades, Country B will be much richer than Country A, even though the growth rate difference seems small.

Rule of 70: Classroom Example

- If Country A grows at 2% per year:
  - It takes about $70/2 = 35$ years to double income per person.
- If Country B grows at 4% per year:
  - It takes about $70/4 = 17.5$ years to double.
- After a few decades, Country B will be much richer than Country A, even though the growth rate difference seems small.

Rule of 70: Classroom Example

- If Country A grows at 2% per year:
  - It takes about $70/2 = 35$ years to double income per person.
- If Country B grows at 4% per year:
  - It takes about $70/4 = 17.5$ years to double.
- After a few decades, Country B will be much richer than Country A, even though the growth rate difference seems small.

Rule of 70: Classroom Example

- If Country A grows at 2% per year:
  - It takes about $70/2 = 35$ years to double income per person.
- If Country B grows at 4% per year:
  - It takes about $70/4 = 17.5$ years to double.
- After a few decades, Country B will be much richer than Country A, even though the growth rate difference seems small.

Rule of 70: Classroom Example

- If Country A grows at 2% per year:
  - It takes about $70/2 = 35$ years to double income per person.
- If Country B grows at 4% per year:
  - It takes about $70/4 = 17.5$ years to double.
- After a few decades, Country B will be much richer than Country A, even though the growth rate difference seems small.

What Determines Long-Run Growth?

- Long-run increases in real GDP per capita depend on increases in **labour productivity**:

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Number of workers or hours worked}}.$$

- The average Canadian today consumes more because:
  - the average Canadian **produces** much more per hour than in 1961.
- So, what makes workers more productive?

What Determines Long-Run Growth?

- Long-run increases in real GDP per capita depend on increases in **labour productivity**:

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Number of workers or hours worked}}.$$

- The average Canadian today consumes more because:
  - the average Canadian **produces** much more per hour than in 1961.
- So, what makes workers more productive?

What Determines Long-Run Growth?

- Long-run increases in real GDP per capita depend on increases in **labour productivity**:

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Number of workers or hours worked}}.$$

- The average Canadian today consumes more because:
  - the average Canadian **produces** much more per hour than in 1961.
- So, what makes workers more productive?

What Determines Long-Run Growth?

- Long-run increases in real GDP per capita depend on increases in **labour productivity**:

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Number of workers or hours worked}}.$$

- The average Canadian today consumes more because:
    - the average Canadian **produces** much more per hour than in 1961.
- So, what makes workers more productive?

Factors Affecting Labour Productivity Growth

- Two key factors:
  - Increases in capital per hour worked
  - Technological change

Factors Affecting Labour Productivity Growth

- Two key factors:
  - **Increases in capital per hour worked**
  - **Technological change**

Factors Affecting Labour Productivity Growth

- Two key factors:
  - **Increases in capital per hour worked**
  - **Technological change**

Capital per Hour Worked

- **Capital:** durable manufactured goods used to produce other goods and services.
  - Machines, computers, tools, buildings, vehicles.
- The more capital each worker has, the more they can produce per hour.
- Also important: **human capital**:
  - The accumulated knowledge and skills workers possess (education, training, experience).
- Example:
  - A worker with a modern computer and software is usually more productive than one with only pen and paper.

Capital per Hour Worked

- **Capital:** durable manufactured goods used to produce other goods and services.
  - Machines, computers, tools, buildings, vehicles.
- The more capital each worker has, the more they can produce per hour.
- Also important: **human capital**:
  - The accumulated knowledge and skills workers possess (education, training, experience).
- Example:
  - A worker with a modern computer and software is usually more productive than one with only pen and paper.

Capital per Hour Worked

- **Capital:** durable manufactured goods used to produce other goods and services.
  - Machines, computers, tools, buildings, vehicles.
- The more capital each worker has, the more they can produce per hour.
- Also important: **human capital**:
  - The accumulated knowledge and skills workers possess (education, training, experience).
- Example:
  - A worker with a modern computer and software is usually more productive than one with only pen and paper.

Capital per Hour Worked

- **Capital:** durable manufactured goods used to produce other goods and services.
  - Machines, computers, tools, buildings, vehicles.
- The more capital each worker has, the more they can produce per hour.
- Also important: **human capital**:
  - The accumulated knowledge and skills workers possess (education, training, experience).
- Example:
  - A worker with a modern computer and software is usually more productive than one with only pen and paper.

Capital per Hour Worked

- **Capital:** durable manufactured goods used to produce other goods and services.
    - Machines, computers, tools, buildings, vehicles.
- The more capital each worker has, the more they can produce per hour.
- Also important: **human capital**:
    - The accumulated knowledge and skills workers possess (education, training, experience).
- Example:
    - A worker with a modern computer and software is usually more productive than one with only pen and paper.

Capital per Hour Worked

- **Capital:** durable manufactured goods used to produce other goods and services.
  - Machines, computers, tools, buildings, vehicles.
- The more capital each worker has, the more they can produce per hour.
- Also important: **human capital**:
  - The accumulated knowledge and skills workers possess (education, training, experience).
- Example:
  - A worker with a modern computer and software is usually more productive than one with only pen and paper.

Capital per Hour Worked

- **Capital:** durable manufactured goods used to produce other goods and services.
  - Machines, computers, tools, buildings, vehicles.
- The more capital each worker has, the more they can produce per hour.
- Also important: **human capital**:
  - The accumulated knowledge and skills workers possess (education, training, experience).
- Example:
  - A worker with a modern computer and software is usually more productive than one with only pen and paper.

Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes,
  - Better management techniques,
  - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
  - They pioneer new ways to bring together labour, capital, and ideas.

Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes,
  - Better management techniques,
  - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
  - They pioneer new ways to bring together labour, capital, and ideas.

Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes,
  - Better management techniques,
  - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
  - They pioneer new ways to bring together labour, capital, and ideas.

Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes,
  - Better management techniques,
  - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
  - They pioneer new ways to bring together labour, capital, and ideas.

Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes,
  - Better management techniques,
  - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
  - They pioneer new ways to bring together labour, capital, and ideas.

Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes,
  - Better management techniques,
  - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
  - They pioneer new ways to bring together labour, capital, and ideas.

Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes,
  - Better management techniques,
  - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
  - They pioneer new ways to bring together labour, capital, and ideas.

Technological Change

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
  - New production processes,
  - Better management techniques,
  - New products (e.g., smartphones, advanced medical devices).
- With better technology, workers can produce more **with the same amount** of capital and labour.
- **Entrepreneurs** play a critical role:
  - They pioneer new ways to bring together labour, capital, and ideas.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

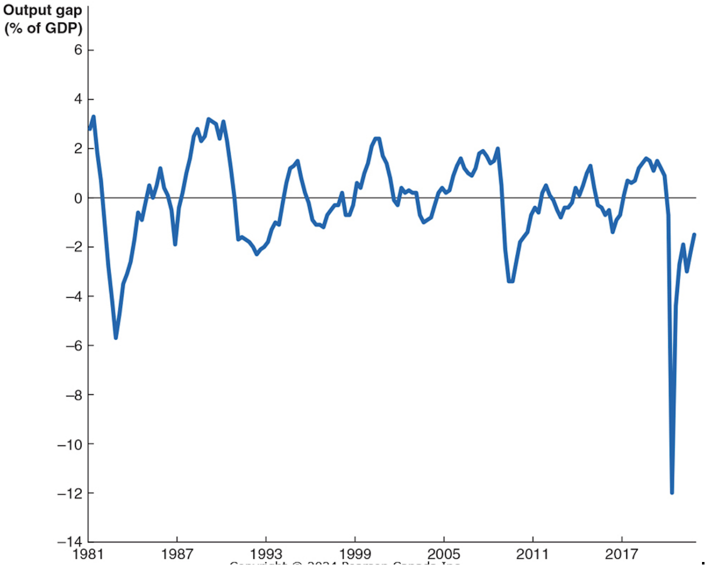
Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Potential GDP

- **Potential GDP:** level of real GDP attained when all firms are producing at **capacity**.
- Capacity here means:
  - "Normal" hours,
  - "Normal" sized workforce,
  - No special overtime or emergency use of capital.
- Potential GDP increases when:
  - labour force grows,
  - more factories are built,
  - more equipment is installed,
  - technological change occurs.

Figure 6.2 – The Output Gap



The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
  - actual GDP is above potential,
  - the economy is using resources in an unsustainable way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
  - actual GDP is below potential,
  - resources (labour and capital) are underutilized.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
  - actual GDP is above potential,
  - the economy is using resources in an unsustainable way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
  - actual GDP is below potential,
  - resources (labour and capital) are underutilized.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
  - actual GDP is above potential,
  - the economy is using resources in an unsustainable way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
  - actual GDP is below potential,
  - resources (labour and capital) are underutilized.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
  - actual GDP is above potential,
  - the economy is using resources in an unsustainable way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
  - actual GDP is below potential,
  - resources (labour and capital) are underutilized.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
  - actual GDP is above potential,
  - the economy is using resources in an unsustainable way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
  - actual GDP is below potential,
  - resources (labour and capital) are underutilized.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
  - actual GDP is above potential,
  - the economy is using resources in an unsustainable way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
  - actual GDP is below potential,
  - resources (labour and capital) are underutilized.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
  - actual GDP is above potential,
  - the economy is using resources in an unsustainable way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
  - actual GDP is below potential,
  - resources (labour and capital) are underutilized.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

The Output Gap

- **Output gap:** difference between actual GDP and potential GDP.
- When the output gap is **positive**:
  - actual GDP is above potential,
  - the economy is using resources in an unsustainable way (very low unemployment, high capacity usage).
- When the output gap is **negative**:
  - actual GDP is below potential,
  - resources (labour and capital) are underutilized.
- The Bank of Canada pays close attention to the output gap when setting interest rates.

6.2 Learning Objective

Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
  - more capital per worker, and
  - ongoing technological improvements.
- Both require **investment**.
- Investment is financed by **saving**, and the financial system connects savers and borrowers.

6.2 Learning Objective

Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
  - more capital per worker, and
  - ongoing technological improvements.
- Both require **investment**.
- Investment is financed by **saving**, and the financial system connects savers and borrowers.

6.2 Learning Objective

Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
  - more capital per worker, and
  - ongoing technological improvements.
- Both require **investment**.
- Investment is financed by **saving**, and the financial system connects savers and borrowers.

6.2 Learning Objective

Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
  - more capital per worker, and
  - ongoing technological improvements.
- Both require **investment**.
- Investment is financed by **saving**, and the financial system connects savers and borrowers.

6.2 Learning Objective

Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
  - more capital per worker, and
  - ongoing technological improvements.
- Both require **investment**.
- Investment is financed by **saving**, and the financial system connects savers and borrowers.

An Overview of the Financial System

- **Financial markets:** markets where financial securities are bought and sold.
  - Examples: stock markets, bond markets.
- **Financial security:** document (often electronic) stating terms under which funds pass from buyer to seller.
- **Stock:** financial security representing partial ownership of a firm.
- **Bond:** financial security promising to repay a fixed amount of funds.
  - In effect, a bond is a **loan** from households to firms or governments.

An Overview of the Financial System

- **Financial markets:** markets where financial securities are bought and sold.
  - Examples: stock markets, bond markets.
- **Financial security:** document (often electronic) stating terms under which funds pass from buyer to seller.
- **Stock:** financial security representing partial ownership of a firm.
- **Bond:** financial security promising to repay a fixed amount of funds.
  - In effect, a bond is a **loan** from households to firms or governments.

An Overview of the Financial System

- **Financial markets:** markets where financial securities are bought and sold.
  - Examples: stock markets, bond markets.
- **Financial security:** document (often electronic) stating terms under which funds pass from buyer to seller.
- **Stock:** financial security representing partial ownership of a firm.
- **Bond:** financial security promising to repay a fixed amount of funds.
  - In effect, a bond is a **loan** from households to firms or governments.

An Overview of the Financial System

- **Financial markets:** markets where financial securities are bought and sold.
  - Examples: stock markets, bond markets.
- **Financial security:** document (often electronic) stating terms under which funds pass from buyer to seller.
- **Stock:** financial security representing partial ownership of a firm.
- **Bond:** financial security promising to repay a fixed amount of funds.
  - In effect, a bond is a **loan** from households to firms or governments.

An Overview of the Financial System

- **Financial markets:** markets where financial securities are bought and sold.
  - Examples: stock markets, bond markets.
- **Financial security:** document (often electronic) stating terms under which funds pass from buyer to seller.
- **Stock:** financial security representing partial ownership of a firm.
- **Bond:** financial security promising to repay a fixed amount of funds.
  - In effect, a bond is a **loan** from households to firms or governments.

An Overview of the Financial System

- **Financial markets:** markets where financial securities are bought and sold.
  - Examples: stock markets, bond markets.
- **Financial security:** document (often electronic) stating terms under which funds pass from buyer to seller.
- **Stock:** financial security representing partial ownership of a firm.
- **Bond:** financial security promising to repay a fixed amount of funds.
  - In effect, a bond is a **loan** from households to firms or governments.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Financial Intermediaries

- **Financial intermediaries:** firms that borrow funds from savers and lend them to borrowers.
- Examples:
  - banks,
  - mutual funds,
  - pension funds,
  - insurance companies.
- Example:
  - Mutual funds sell shares to savers,
  - then use the funds to buy a diversified portfolio of stocks and bonds.

Three Key Services of the Financial System

- The financial system provides three key services:
    1. **Risk-sharing**
    2. **Liquidity**
    3. **Information**

Three Key Services of the Financial System

- The financial system provides three key services:
  1. **Risk-sharing**
  2. **Liquidity**
  3. **Information**

Three Key Services of the Financial System

- The financial system provides three key services:
  1. **Risk-sharing**
  2. **Liquidity**
  3. **Information**

Three Key Services of the Financial System

- The financial system provides three key services:
    1. **Risk-sharing**
    2. **Liquidity**
    3. **Information**

Risk-Sharing

- **Risk-sharing:**
  - Savers can spread their money over many different assets.
  - This reduces risk while maintaining a high expected return.
- Example:
  - Buying shares in a mutual fund instead of investing all savings in one firm.

Risk-Sharing

- **Risk-sharing:**
  - Savers can spread their money over many different assets.
  - This reduces risk while maintaining a high expected return.
- Example:
  - Buying shares in a mutual fund instead of investing all savings in one firm.

Risk-Sharing

- **Risk-sharing:**
  - Savers can spread their money over many different assets.
  - This reduces risk while maintaining a high expected return.
- Example:
  - Buying shares in a mutual fund instead of investing all savings in one firm.

Risk-Sharing

- **Risk-sharing:**
  - Savers can spread their money over many different assets.
  - This reduces risk while maintaining a high expected return.
- Example:
  - Buying shares in a mutual fund instead of investing all savings in one firm.

Risk-Sharing

- **Risk-sharing:**
  - Savers can spread their money over many different assets.
  - This reduces risk while maintaining a high expected return.
- Example:
  - Buying shares in a mutual fund instead of investing all savings in one firm.

Liquidity

- **Liquidity:** the ease with which a financial asset can be converted into cash.
- The financial system provides liquid assets:
  - bank accounts,
  - money market mutual funds,
  - actively traded bonds and stocks.
- This allows savers to commit funds while still being able to access cash when needed.

Liquidity

- **Liquidity:** the ease with which a financial asset can be converted into cash.
- The financial system provides liquid assets:
  - bank accounts,
  - money market mutual funds,
  - actively traded bonds and stocks.
- This allows savers to commit funds while still being able to access cash when needed.

Liquidity

- **Liquidity:** the ease with which a financial asset can be converted into cash.
- The financial system provides liquid assets:
  - bank accounts,
  - money market mutual funds,
  - actively traded bonds and stocks.
- This allows savers to commit funds while still being able to access cash when needed.

Liquidity

- **Liquidity:** the ease with which a financial asset can be converted into cash.
- The financial system provides liquid assets:
  - bank accounts,
  - money market mutual funds,
  - actively traded bonds and stocks.
- This allows savers to commit funds while still being able to access cash when needed.

Liquidity

- **Liquidity:** the ease with which a financial asset can be converted into cash.
- The financial system provides liquid assets:
  - bank accounts,
  - money market mutual funds,
  - actively traded bonds and stocks.
- This allows savers to commit funds while still being able to access cash when needed.

Liquidity

- **Liquidity:** the ease with which a financial asset can be converted into cash.
- The financial system provides liquid assets:
  - bank accounts,
  - money market mutual funds,
  - actively traded bonds and stocks.
- This allows savers to commit funds while still being able to access cash when needed.

Information

- **Information:**
  - Prices of financial securities reflect the expectations of many investors about future returns.
- The financial system aggregates information about:
  - firm profitability,
  - riskiness of projects,
  - macroeconomic conditions.
- This guides funds toward the most promising investment opportunities.

Information

- **Information:**
  - Prices of financial securities reflect the expectations of many investors about future returns.
- The financial system aggregates information about:
  - firm profitability,
  - riskiness of projects,
  - macroeconomic conditions.
- This guides funds toward the most promising investment opportunities.

Information

- **Information:**
  - Prices of financial securities reflect the expectations of many investors about future returns.
- The financial system aggregates information about:
  - firm profitability,
  - riskiness of projects,
  - macroeconomic conditions.
- This guides funds toward the most promising investment opportunities.

Information

- **Information:**
  - Prices of financial securities reflect the expectations of many investors about future returns.
- The financial system aggregates information about:
  - firm profitability,
  - riskiness of projects,
  - macroeconomic conditions.
- This guides funds toward the most promising investment opportunities.

Information

- **Information:**
  - Prices of financial securities reflect the expectations of many investors about future returns.
- The financial system aggregates information about:
  - firm profitability,
  - riskiness of projects,
  - macroeconomic conditions.
- This guides funds toward the most promising investment opportunities.

Information

- **Information:**
    - Prices of financial securities reflect the expectations of many investors about future returns.
- The financial system aggregates information about:
    - firm profitability,
    - riskiness of projects,
    - macroeconomic conditions.
- This guides funds toward the most promising investment opportunities.

Information

- **Information:**
  - Prices of financial securities reflect the expectations of many investors about future returns.
- The financial system aggregates information about:
  - firm profitability,
  - riskiness of projects,
  - macroeconomic conditions.
- This guides funds toward the most promising investment opportunities.

The Macroeconomics of Saving and Investment

- Start from the national income identity:

$$Y = C + I + G + NX.$$

- For simplicity, assume a **closed economy** (no exports or imports), so:

$$Y = C + I + G.$$

- Rearranging to solve for investment:

$$I = Y - C - G.$$

- Investment equals total income not used for consumption or government purchases.

The Macroeconomics of Saving and Investment

- Start from the national income identity:

$$Y = C + I + G + NX.$$

- For simplicity, assume a **closed economy** (no exports or imports), so:

$$Y = C + I + G.$$

- Rearranging to solve for investment:

$$I = Y - C - G.$$

- Investment equals total income not used for consumption or government purchases.

The Macroeconomics of Saving and Investment

- Start from the national income identity:

$$Y = C + I + G + NX.$$

- For simplicity, assume a **closed economy** (no exports or imports), so:

$$Y = C + I + G.$$

- Rearranging to solve for investment:

$$I = Y - C - G.$$

- Investment equals total income not used for consumption or government purchases.

The Macroeconomics of Saving and Investment

- Start from the national income identity:

$$Y = C + I + G + NX.$$

- For simplicity, assume a **closed economy** (no exports or imports), so:

$$Y = C + I + G.$$

- Rearranging to solve for investment:

$$I = Y - C - G.$$

- Investment equals total income not used for consumption or government purchases.

Private Saving and Public Saving

- **Private saving** (S_{private}):
  - Households receive income from factors of production (Y) and transfers (TR).
  - They pay taxes (T) and consume (C).
- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving** (S_{public}):
  - The government receives tax revenue (T),
  - and spends on purchases (G) and transfers (TR).
- So:

$$S_{\text{public}} = T - G - \text{TR}.$$

Private Saving and Public Saving

- **Private saving** (S_{private}):
  - Households receive income from factors of production (Y) and transfers (TR).
  - They pay taxes (T) and consume (C).
- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving** (S_{public}):
  - The government receives tax revenue (T),
  - and spends on purchases (G) and transfers (TR).
- So:

$$S_{\text{public}} = T - G - \text{TR}.$$

Private Saving and Public Saving

- **Private saving** (S_{private}):
  - Households receive income from factors of production (Y) and transfers (TR).
  - They pay taxes (T) and consume (C).
- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving** (S_{public}):
  - The government receives tax revenue (T),
  - and spends on purchases (G) and transfers (TR).
- So:

$$S_{\text{public}} = T - G - \text{TR}.$$

Private Saving and Public Saving

- **Private saving** (S_{private}):
  - Households receive income from factors of production (Y) and transfers (TR).
  - They pay taxes (T) and consume (C).
- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving** (S_{public}):
  - The government receives tax revenue (T),
  - and spends on purchases (G) and transfers (TR).
- So:

$$S_{\text{public}} = T - G - \text{TR}.$$

Private Saving and Public Saving

- **Private saving** (S_{private}):
  - Households receive income from factors of production (Y) and transfers (TR).
  - They pay taxes (T) and consume (C).
- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving** (S_{public}):
  - The government receives tax revenue (T),
  - and spends on purchases (G) and transfers (TR).
- So:

$$S_{\text{public}} = T - G - \text{TR}.$$

Private Saving and Public Saving

- **Private saving** (S_{private}):
  - Households receive income from factors of production (Y) and transfers (TR).
  - They pay taxes (T) and consume (C).
- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving** (S_{public}):
  - The government receives tax revenue (T),
  - and spends on purchases (G) and transfers (TR).
- So:

$$S_{\text{public}} = T - G - \text{TR}.$$

Private Saving and Public Saving

- **Private saving** (S_{private}):
  - Households receive income from factors of production (Y) and transfers (TR).
  - They pay taxes (T) and consume (C).
- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving** (S_{public}):
  - The government receives tax revenue (T),
  - and spends on purchases (G) and transfers (TR).
- So:

$$S_{\text{public}} = T - G - \text{TR}.$$

Private Saving and Public Saving

- **Private saving** (S_{private}):
  - Households receive income from factors of production (Y) and transfers (TR).
  - They pay taxes (T) and consume (C).
- So:

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving** (S_{public}):
  - The government receives tax revenue (T),
  - and spends on purchases (G) and transfers (TR).
- So:

$$S_{\text{public}} = T - G - \text{TR}.$$

Total Saving and Investment

- **Total saving** is:

$$S = S_{\text{private}} + S_{\text{public}}.$$

- Substitute the formulas:

$$\begin{aligned} S &= (Y + \text{TR} - C - T) + (T - G - \text{TR}) \\ &= Y - C - G. \end{aligned}$$

- But from earlier:

$$I = Y - C - G.$$

- Therefore:

$$S = I.$$

- In a closed economy, **saving equals investment**.

Total Saving and Investment

- **Total saving** is:

$$S = S_{\text{private}} + S_{\text{public}}.$$

- Substitute the formulas:

$$\begin{aligned} S &= (Y + \text{TR} - C - T) + (T - G - \text{TR}) \\ &= Y - C - G. \end{aligned}$$

- But from earlier:

$$I = Y - C - G.$$

- Therefore:

$$S = I.$$

- In a closed economy, **saving equals investment**.

Total Saving and Investment

- **Total saving** is:

$$S = S_{\text{private}} + S_{\text{public}}.$$

- Substitute the formulas:

$$\begin{aligned} S &= (Y + \mathrm{TR} - C - T) + (T - G - \mathrm{TR}) \\ &= Y - C - G. \end{aligned}$$

- But from earlier:

$$I = Y - C - G.$$

- Therefore:

$$S = I.$$

- In a closed economy, **saving equals investment**.

Total Saving and Investment

- **Total saving** is:

$$S = S_{\text{private}} + S_{\text{public}}.$$

- Substitute the formulas:

$$\begin{aligned} S &= (Y + \text{TR} - C - T) + (T - G - \text{TR}) \\ &= Y - C - G. \end{aligned}$$

- But from earlier:

$$I = Y - C - G.$$

- Therefore:

$$S = I.$$

- In a closed economy, **saving equals investment.**

Total Saving and Investment

- **Total saving** is:
$$S = S_{\text{private}} + S_{\text{public}}.$$
- Substitute the formulas:
$$\begin{aligned} S &= (Y + \mathsf{TR} - C - T) + (T - G - \mathsf{TR}) \\ &= Y - C - G. \end{aligned}$$
- But from earlier:
$$I = Y - C - G.$$
- Therefore:
$$S = I.$$
- In a closed economy, **saving equals investment**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

Budget Balance and Crowding Out

- If $S_{\text{public}} = 0$:
  - Government has a **balanced budget**.
- If $S_{\text{public}} < 0$:
  - Government runs a **budget deficit**.
  - It must borrow (sell bonds), reducing funds available for private investment.
- If $S_{\text{public}} > 0$:
  - Government runs a **budget surplus**.
  - It contributes to national saving.
- When deficits reduce investment, we call this **crowding out**.

The Market for Loanable Funds

- We can think of all financial markets together as one **market for loanable funds**.
- **Borrowers:** firms that want to invest in capital (and the government when it runs deficits).
- **Lenders:** households that save and lend funds, and sometimes foreign investors.
- The **real interest rate** is the price that equates saving and investment.

The Market for Loanable Funds

- We can think of all financial markets together as one **market for loanable funds**.
- **Borrowers:** firms that want to invest in capital (and the government when it runs deficits).
- **Lenders:** households that save and lend funds, and sometimes foreign investors.
- The **real interest rate** is the price that equates saving and investment.

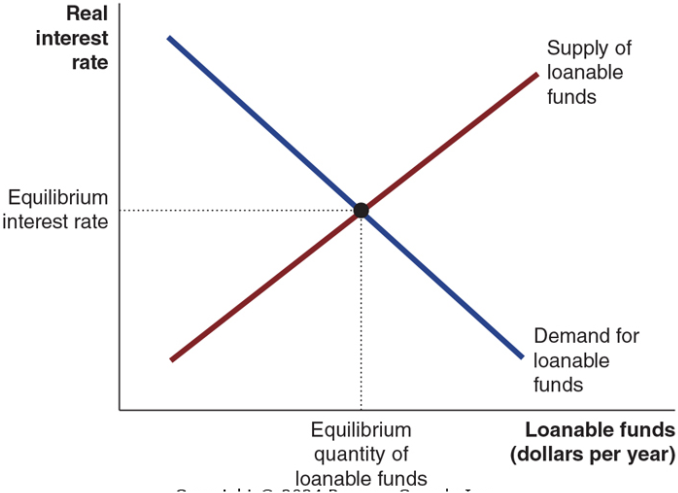
The Market for Loanable Funds

- We can think of all financial markets together as one **market for loanable funds**.
- **Borrowers:** firms that want to invest in capital (and the government when it runs deficits).
- **Lenders:** households that save and lend funds, and sometimes foreign investors.
- The **real interest rate** is the price that equates saving and investment.

The Market for Loanable Funds

- We can think of all financial markets together as one **market for loanable funds**.
- **Borrowers:** firms that want to invest in capital (and the government when it runs deficits).
- **Lenders:** households that save and lend funds, and sometimes foreign investors.
- The **real interest rate** is the price that equates saving and investment.

Figure 6.3 – The Market for Loanable Funds



Supply and Demand for Loanable Funds

- **Demand for loanable funds:**
  - Firms borrow more when the real interest rate is lower,
  - because more investment projects are profitable.
- **Supply of loanable funds:**
  - Households save more when the real interest rate is higher,
  - because the reward for postponing consumption is greater.
- Government saving (or dissaving) also affects supply.

Supply and Demand for Loanable Funds

- **Demand for loanable funds:**
  - Firms borrow more when the real interest rate is lower,
  - because more investment projects are profitable.
- **Supply of loanable funds:**
  - Households save more when the real interest rate is higher,
  - because the reward for postponing consumption is greater.
- Government saving (or dissaving) also affects supply.

Supply and Demand for Loanable Funds

- **Demand for loanable funds:**
  - Firms borrow more when the real interest rate is lower,
  - because more investment projects are profitable.
- Supply of loanable funds:
  - Households save more when the real interest rate is higher,
  - because the reward for postponing consumption is greater.
- Government saving (or dissaving) also affects supply.

Supply and Demand for Loanable Funds

- **Demand for loanable funds:**
  - Firms borrow more when the real interest rate is lower,
  - because more investment projects are profitable.
- **Supply of loanable funds:**
  - Households save more when the real interest rate is higher,
  - because the reward for postponing consumption is greater.
- Government saving (or dissaving) also affects supply.

Supply and Demand for Loanable Funds

- **Demand for loanable funds:**
  - Firms borrow more when the real interest rate is lower,
  - because more investment projects are profitable.
- **Supply of loanable funds:**
  - Households save more when the real interest rate is higher,
  - because the reward for postponing consumption is greater.
- Government saving (or dissaving) also affects supply.

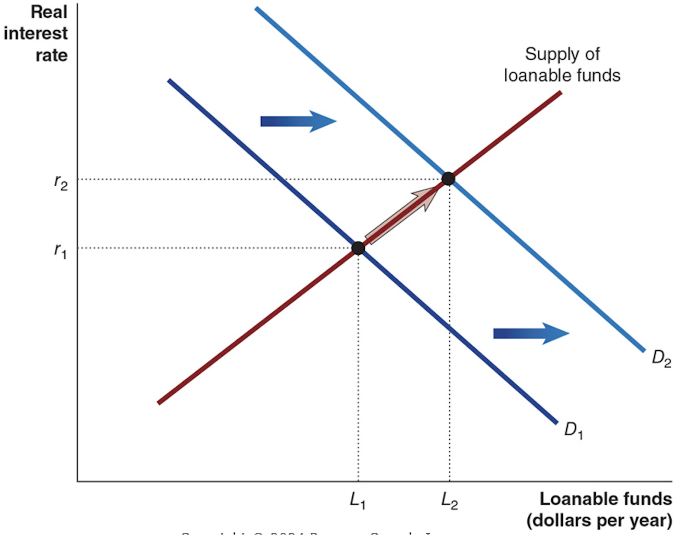
Supply and Demand for Loanable Funds

- **Demand for loanable funds:**
  - Firms borrow more when the real interest rate is lower,
  - because more investment projects are profitable.
- **Supply of loanable funds:**
  - Households save more when the real interest rate is higher,
  - because the reward for postponing consumption is greater.
- Government saving (or dissaving) also affects supply.

Supply and Demand for Loanable Funds

- **Demand for loanable funds:**
  - Firms borrow more when the real interest rate is lower,
  - because more investment projects are profitable.
- **Supply of loanable funds:**
  - Households save more when the real interest rate is higher,
  - because the reward for postponing consumption is greater.
- Government saving (or dissaving) also affects supply.

Figure 6.4 – Increase in Demand for Loanable Funds



Increase in Demand for Loanable Funds

- Suppose technological change makes investment more profitable.
- Firms want to borrow more at each interest rate.
- **Result:**
  - Demand for loanable funds shifts right.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of loanable funds (investment) rises.

Increase in Demand for Loanable Funds

- Suppose technological change makes investment more profitable.
- Firms want to borrow more at each interest rate.
- **Result:**
    - Demand for loanable funds shifts right.
    - Equilibrium real interest rate rises.
    - Equilibrium quantity of loanable funds (investment) rises.

Increase in Demand for Loanable Funds

- Suppose technological change makes investment more profitable.
- Firms want to borrow more at each interest rate.
- **Result:**
    - Demand for loanable funds shifts right.
    - Equilibrium real interest rate rises.
    - Equilibrium quantity of loanable funds (investment) rises.

Increase in Demand for Loanable Funds

- Suppose technological change makes investment more profitable.
- Firms want to borrow more at each interest rate.
- **Result:**
  - Demand for loanable funds shifts right.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of loanable funds (investment) rises.

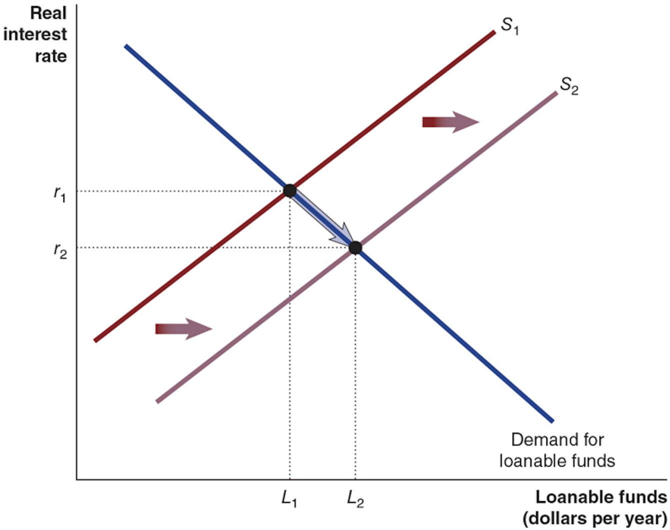
Increase in Demand for Loanable Funds

- Suppose technological change makes investment more profitable.
- Firms want to borrow more at each interest rate.
- **Result:**
  - Demand for loanable funds shifts right.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of loanable funds (investment) rises.

Increase in Demand for Loanable Funds

- Suppose technological change makes investment more profitable.
- Firms want to borrow more at each interest rate.
- **Result:**
  - Demand for loanable funds shifts right.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of loanable funds (investment) rises.

Figure 6.5 – Increase in Supply of Loanable Funds



Increase in Supply of Loanable Funds

- Suppose households become more patient or more concerned about retirement, and decide to save more at every interest rate.
- **Result:**
  - Supply of loanable funds shifts right.
  - Equilibrium real interest rate falls.
  - Equilibrium quantity of loanable funds (investment) rises.

Increase in Supply of Loanable Funds

- Suppose households become more patient or more concerned about retirement, and decide to save more at every interest rate.
- **Result:**
  - Supply of loanable funds shifts right.
  - Equilibrium real interest rate falls.
  - Equilibrium quantity of loanable funds (investment) rises.

Increase in Supply of Loanable Funds

- Suppose households become more patient or more concerned about retirement, and decide to save more at every interest rate.
- **Result:**
  - Supply of loanable funds shifts right.
  - Equilibrium real interest rate falls.
  - Equilibrium quantity of loanable funds (investment) rises.

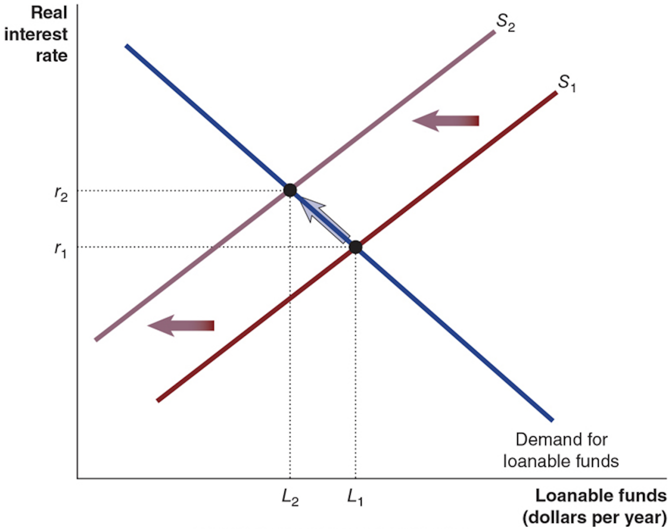
Increase in Supply of Loanable Funds

- Suppose households become more patient or more concerned about retirement, and decide to save more at every interest rate.
- **Result:**
  - Supply of loanable funds shifts right.
  - Equilibrium real interest rate falls.
  - Equilibrium quantity of loanable funds (investment) rises.

Increase in Supply of Loanable Funds

- Suppose households become more patient or more concerned about retirement, and decide to save more at every interest rate.
- **Result:**
  - Supply of loanable funds shifts right.
  - Equilibrium real interest rate falls.
  - Equilibrium quantity of loanable funds (investment) rises.

Figure 6.6 – Effect of a Budget Deficit



Budget Deficits and Crowding Out

- When government runs a **budget deficit**, it must borrow funds.
- This reduces the funds available for private borrowers.
- In the loanable funds market:
  - Supply of loanable funds shifts left.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of funds to firms falls.
- This reduction in private investment due to government borrowing is **crowding out**.

Budget Deficits and Crowding Out

- When government runs a **budget deficit**, it must borrow funds.
- This reduces the funds available for private borrowers.
- In the loanable funds market:
  - Supply of loanable funds shifts left.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of funds to firms falls.
- This reduction in private investment due to government borrowing is **crowding out**.

Budget Deficits and Crowding Out

- When government runs a **budget deficit**, it must borrow funds.
- This reduces the funds available for private borrowers.
- In the loanable funds market:
  - Supply of loanable funds shifts left.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of funds to firms falls.
- This reduction in private investment due to government borrowing is **crowding out**.

Budget Deficits and Crowding Out

- When government runs a **budget deficit**, it must borrow funds.
- This reduces the funds available for private borrowers.
- In the loanable funds market:
  - Supply of loanable funds shifts left.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of funds to firms falls.
- This reduction in private investment due to government borrowing is **crowding out**.

Budget Deficits and Crowding Out

- When government runs a **budget deficit**, it must borrow funds.
- This reduces the funds available for private borrowers.
- In the loanable funds market:
  - Supply of loanable funds shifts left.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of funds to firms falls.
- This reduction in private investment due to government borrowing is **crowding out**.

Budget Deficits and Crowding Out

- When government runs a **budget deficit**, it must borrow funds.
- This reduces the funds available for private borrowers.
- In the loanable funds market:
  - Supply of loanable funds shifts left.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of funds to firms falls.
- This reduction in private investment due to government borrowing is **crowding out**.

Budget Deficits and Crowding Out

- When government runs a **budget deficit**, it must borrow funds.
- This reduces the funds available for private borrowers.
- In the loanable funds market:
  - Supply of loanable funds shifts left.
  - Equilibrium real interest rate rises.
  - Equilibrium quantity of funds to firms falls.
- This reduction in private investment due to government borrowing is **crowding out**.

How Important Is Crowding Out?

- In practice, the effect of deficits on interest rates is often **small**.
- Why?
  - Interest rates are influenced by **global saving**.
  - Even several billion dollars of extra borrowing may be small relative to the world market.
- Small does not mean zero:
  - Higher public debt may require higher taxes later.
  - This can depress long-run growth.

How Important Is Crowding Out?

- In practice, the effect of deficits on interest rates is often **small**.
- Why?
  - Interest rates are influenced by **global saving**.
  - Even several billion dollars of extra borrowing may be small relative to the world market.
- Small does not mean zero:
  - Higher public debt may require higher taxes later.
  - This can depress long-run growth.

How Important Is Crowding Out?

- In practice, the effect of deficits on interest rates is often **small**.
- Why?
  - Interest rates are influenced by **global saving**.
  - Even several billion dollars of extra borrowing may be small relative to the world market.
- Small does not mean zero:
  - Higher public debt may require higher taxes later.
  - This can depress long-run growth.

How Important Is Crowding Out?

- In practice, the effect of deficits on interest rates is often **small**.
- Why?
  - Interest rates are influenced by **global saving**.
  - Even several billion dollars of extra borrowing may be small relative to the world market.
- Small does not mean zero:
  - Higher public debt may require higher taxes later.
  - This can depress long-run growth.

How Important Is Crowding Out?

- In practice, the effect of deficits on interest rates is often **small**.
- Why?
  - Interest rates are influenced by **global saving**.
  - Even several billion dollars of extra borrowing may be small relative to the world market.
- Small does not mean zero:
  - Higher public debt may require higher taxes later.
  - This can depress long-run growth.

How Important Is Crowding Out?

- In practice, the effect of deficits on interest rates is often **small**.
- Why?
  - Interest rates are influenced by **global saving**.
  - Even several billion dollars of extra borrowing may be small relative to the world market.
- Small does not mean zero:
  - Higher public debt may require higher taxes later.
  - This can depress long-run growth.

How Important Is Crowding Out?

- In practice, the effect of deficits on interest rates is often **small**.
- Why?
  - Interest rates are influenced by **global saving**.
  - Even several billion dollars of extra borrowing may be small relative to the world market.
- Small does not mean zero:
  - Higher public debt may require higher taxes later.
  - This can depress long-run growth.

6.3 Learning Objective

Goal

Explain what happens during the business cycle and how it affects inflation and unemployment.

- Real GDP per capita has risen about threefold since 1961.
- But it did not rise **smoothly**—there were periods of recession and expansion.

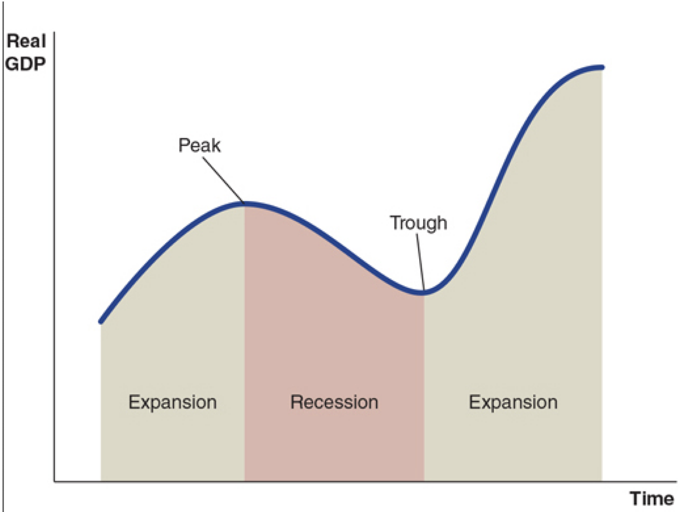
6.3 Learning Objective

Goal

Explain what happens during the business cycle and how it affects inflation and unemployment.

- Real GDP per capita has risen about threefold since 1961.
- But it did not rise **smoothly**—there were periods of recession and expansion.

Figure 6.7 – The Business Cycle (Idealized)



(a) An idealized business cycle

Phases of the Business Cycle

- **Expansion:** real GDP is rising.
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling.
- **Trough:** the point where real GDP stops falling and begins to rise.
- After the trough, a new expansion begins.

Phases of the Business Cycle

- **Expansion:** real GDP is rising.
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling.
- **Trough:** the point where real GDP stops falling and begins to rise.
- After the trough, a new expansion begins.

Phases of the Business Cycle

- **Expansion:** real GDP is rising.
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling.
- **Trough:** the point where real GDP stops falling and begins to rise.
- After the trough, a new expansion begins.

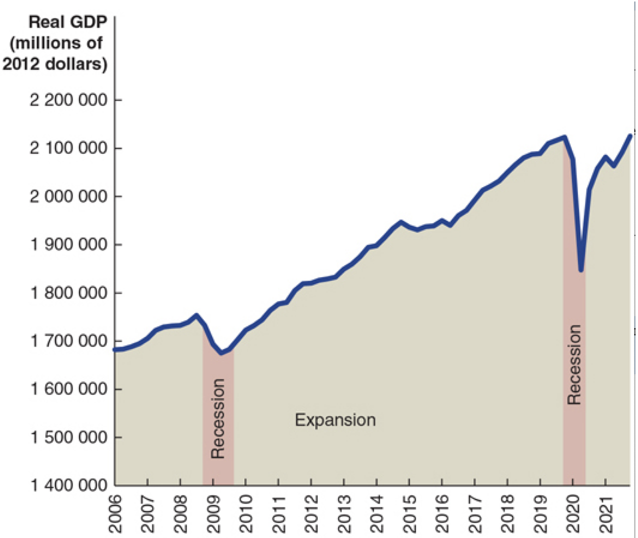
Phases of the Business Cycle

- **Expansion:** real GDP is rising.
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling.
- **Trough:** the point where real GDP stops falling and begins to rise.
- After the trough, a new expansion begins.

Phases of the Business Cycle

- **Expansion:** real GDP is rising.
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling.
- **Trough:** the point where real GDP stops falling and begins to rise.
- After the trough, a new expansion begins.

Figure 6.7 – Canadian Real GDP, 2006–2021



(b) Actual movements in GDP

Recent Business Cycles in Canada

- After a peak in 2008:
  - The economy entered a recession from 2008 to 2009.
  - Real GDP fell, and unemployment rose.
- After the next peak in 2019:
  - A particularly deep recession occurred during 2020 due to the COVID-19 pandemic.
  - The trough occurred between April and June 2020.
  - Real GDP only returned to its 2019 level at the end of 2021.

Recent Business Cycles in Canada

- After a peak in 2008:
  - The economy entered a recession from 2008 to 2009.
  - Real GDP fell, and unemployment rose.
- After the next peak in 2019:
  - A particularly deep recession occurred during 2020 due to the COVID-19 pandemic.
  - The trough occurred between April and June 2020.
  - Real GDP only returned to its 2019 level at the end of 2021.

Recent Business Cycles in Canada

- After a peak in 2008:
  - The economy entered a recession from 2008 to 2009.
  - Real GDP fell, and unemployment rose.
- After the next peak in 2019:
  - A particularly deep recession occurred during 2020 due to the COVID-19 pandemic.
  - The trough occurred between April and June 2020.
  - Real GDP only returned to its 2019 level at the end of 2021.

Recent Business Cycles in Canada

- After a peak in 2008:
  - The economy entered a recession from 2008 to 2009.
  - Real GDP fell, and unemployment rose.
- After the next peak in 2019:
  - A particularly deep recession occurred during 2020 due to the COVID-19 pandemic.
  - The trough occurred between April and June 2020.
  - Real GDP only returned to its 2019 level at the end of 2021.

Recent Business Cycles in Canada

- After a peak in 2008:
  - The economy entered a recession from 2008 to 2009.
  - Real GDP fell, and unemployment rose.
- After the next peak in 2019:
  - A particularly deep recession occurred during 2020 due to the COVID-19 pandemic.
  - The trough occurred between April and June 2020.
  - Real GDP only returned to its 2019 level at the end of 2021.

Recent Business Cycles in Canada

- After a peak in 2008:
  - The economy entered a recession from 2008 to 2009.
  - Real GDP fell, and unemployment rose.
- After the next peak in 2019:
  - A particularly deep recession occurred during 2020 due to the COVID-19 pandemic.
  - The trough occurred between April and June 2020.
  - Real GDP only returned to its 2019 level at the end of 2021.

Recent Business Cycles in Canada

- After a peak in 2008:
  - The economy entered a recession from 2008 to 2009.
  - Real GDP fell, and unemployment rose.
- After the next peak in 2019:
  - A particularly deep recession occurred during 2020 due to the COVID-19 pandemic.
  - The trough occurred between April and June 2020.
  - Real GDP only returned to its 2019 level at the end of 2021.

How Do We Know the Economy Is in a Recession?

- Statistics Canada provides many data series (real GDP, employment, income, etc.).
- There is no single Canadian agency that officially declares recessions.
- Common rule of thumb:
    - A recession is two consecutive quarters of negative growth in real GDP.
    - That is six months of falling real GDP.

How Do We Know the Economy Is in a Recession?

- Statistics Canada provides many data series (real GDP, employment, income, etc.).
- There is no single Canadian agency that officially declares recessions.
- Common rule of thumb:
  - A recession is two consecutive quarters of negative growth in real GDP.
  - That is six months of falling real GDP.

How Do We Know the Economy Is in a Recession?

- Statistics Canada provides many data series (real GDP, employment, income, etc.).
- There is no single Canadian agency that officially declares recessions.
- Common rule of thumb:
  - A recession is two consecutive quarters of negative growth in real GDP.
  - That is six months of falling real GDP.

How Do We Know the Economy Is in a Recession?

- Statistics Canada provides many data series (real GDP, employment, income, etc.).
- There is no single Canadian agency that officially declares recessions.
- Common rule of thumb:
  - A recession is two consecutive quarters of negative growth in real GDP.
  - That is six months of falling real GDP.

How Do We Know the Economy Is in a Recession?

- Statistics Canada provides many data series (real GDP, employment, income, etc.).
- There is no single Canadian agency that officially declares recessions.
- Common rule of thumb:
  - A recession is two consecutive quarters of negative growth in real GDP.
  - That is six months of falling real GDP.

The Effect of the Business Cycle on Inflation

- **Inflation rate:** percentage increase in the price level from one year to the next.
- During expansions:
  - Demand for products is high relative to supply.
  - Firms can raise prices more easily $\Rightarrow$ higher inflation.
- During recessions:
  - Demand for products is low relative to supply.
  - Prices increase more slowly or may even fall $\Rightarrow$ low inflation or deflation.

The Effect of the Business Cycle on Inflation

- **Inflation rate:** percentage increase in the price level from one year to the next.
- During expansions:
  - Demand for products is high relative to supply.
  - Firms can raise prices more easily ⇒ higher inflation.
- During recessions:
  - Demand for products is low relative to supply.
  - Prices increase more slowly or may even fall ⇒ low inflation or deflation.

The Effect of the Business Cycle on Inflation

- **Inflation rate:** percentage increase in the price level from one year to the next.
- During expansions:
  - Demand for products is high relative to supply.
  - Firms can raise prices more easily $\Rightarrow$ higher inflation.
- During recessions:
  - Demand for products is low relative to supply.
  - Prices increase more slowly or may even fall $\Rightarrow$ low inflation or deflation.

The Effect of the Business Cycle on Inflation

- **Inflation rate:** percentage increase in the price level from one year to the next.
- During expansions:
  - Demand for products is high relative to supply.
  - Firms can raise prices more easily $\Rightarrow$ higher inflation.
- During recessions:
  - Demand for products is low relative to supply.
  - Prices increase more slowly or may even fall $\Rightarrow$ low inflation or deflation.

The Effect of the Business Cycle on Inflation

- **Inflation rate:** percentage increase in the price level from one year to the next.
- During expansions:
  - Demand for products is high relative to supply.
  - Firms can raise prices more easily $\Rightarrow$ higher inflation.
- During recessions:
  - Demand for products is low relative to supply.
  - Prices increase more slowly or may even fall $\Rightarrow$ low inflation or deflation.

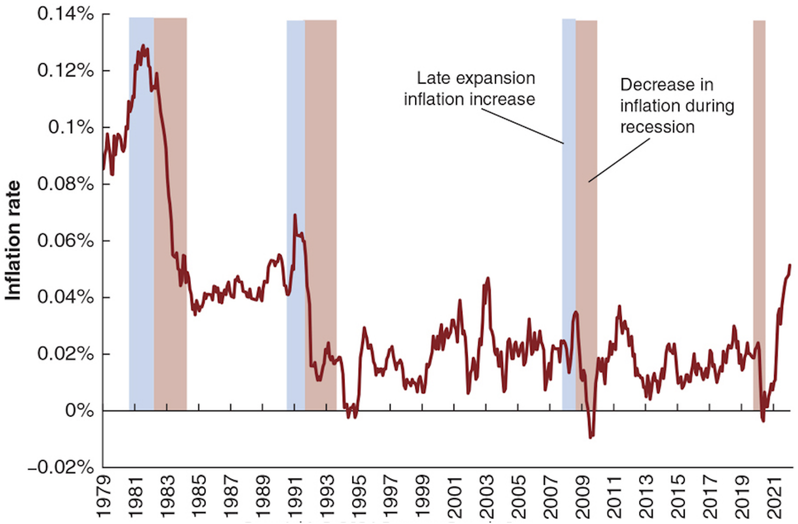
The Effect of the Business Cycle on Inflation

- **Inflation rate:** percentage increase in the price level from one year to the next.
- During expansions:
  - Demand for products is high relative to supply.
  - Firms can raise prices more easily $\Rightarrow$ higher inflation.
- During recessions:
  - Demand for products is low relative to supply.
  - Prices increase more slowly or may even fall $\Rightarrow$ low inflation or deflation.

The Effect of the Business Cycle on Inflation

- **Inflation rate:** percentage increase in the price level from one year to the next.
- During expansions:
  - Demand for products is high relative to supply.
  - Firms can raise prices more easily $\Rightarrow$ higher inflation.
- During recessions:
  - Demand for products is low relative to supply.
  - Prices increase more slowly or may even fall $\Rightarrow$ low inflation or deflation.

Figure 6.8 – Recessions and the Inflation Rate



Inflation Over the Business Cycle

- In each of the last three Canadian recessions:
  - Inflation tended to rise during the preceding expansion.
  - Inflation tended to fall over the course of the recession.
- This pattern helps central banks:
  - use inflation as one indicator of where the economy is in the business cycle.

Inflation Over the Business Cycle

- In each of the last three Canadian recessions:
  - Inflation tended to rise during the preceding expansion.
  - Inflation tended to fall over the course of the recession.
- This pattern helps central banks:
  - use inflation as one indicator of where the economy is in the business cycle.

Inflation Over the Business Cycle

- In each of the last three Canadian recessions:
  - Inflation tended to rise during the preceding expansion.
  - Inflation tended to fall over the course of the recession.
- This pattern helps central banks:
  - use inflation as one indicator of where the economy is in the business cycle.

Inflation Over the Business Cycle

- In each of the last three Canadian recessions:
  - Inflation tended to rise during the preceding expansion.
  - Inflation tended to fall over the course of the recession.
- This pattern helps central banks:
  - use inflation as one indicator of where the economy is in the business cycle.

Inflation Over the Business Cycle

- In each of the last three Canadian recessions:
  - Inflation tended to rise during the preceding expansion.
  - Inflation tended to fall over the course of the recession.
- This pattern helps central banks:
  - use inflation as one indicator of where the economy is in the business cycle.

The Effect of the Business Cycle on Unemployment

- During recessions:
  - Firms see sales fall.
  - They reduce production.
  - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
  - Firms hire more workers.
  - Unemployment gradually falls.

The Effect of the Business Cycle on Unemployment

- During recessions:
  - Firms see sales fall.
  - They reduce production.
  - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
  - Firms hire more workers.
  - Unemployment gradually falls.

The Effect of the Business Cycle on Unemployment

- During recessions:
  - Firms see sales fall.
  - They reduce production.
  - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
  - Firms hire more workers.
  - Unemployment gradually falls.

The Effect of the Business Cycle on Unemployment

- During recessions:
  - Firms see sales fall.
  - They reduce production.
  - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
  - Firms hire more workers.
  - Unemployment gradually falls.

The Effect of the Business Cycle on Unemployment

- During recessions:
  - Firms see sales fall.
  - They reduce production.
  - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
  - Firms hire more workers.
  - Unemployment gradually falls.

The Effect of the Business Cycle on Unemployment

- During recessions:
  - Firms see sales fall.
  - They reduce production.
  - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
  - Firms hire more workers.
  - Unemployment gradually falls.

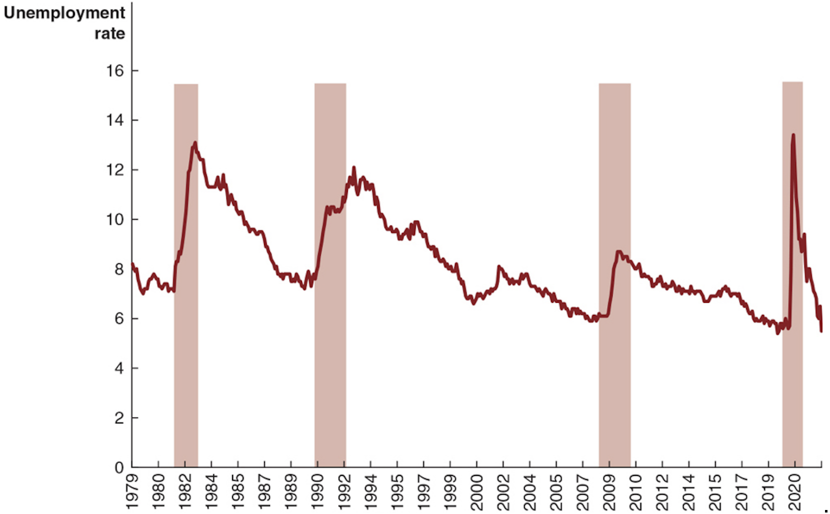
The Effect of the Business Cycle on Unemployment

- During recessions:
  - Firms see sales fall.
  - They reduce production.
  - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
  - Firms hire more workers.
  - Unemployment gradually falls.

The Effect of the Business Cycle on Unemployment

- During recessions:
  - Firms see sales fall.
  - They reduce production.
  - They lay off workers.
- **Unemployment** often continues to rise even after a recession begins.
- During expansions:
  - Firms hire more workers.
  - Unemployment gradually falls.

Figure 6.9 – Recessions and the Unemployment Rate



Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- Financial system:
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Putting It All Together

- **Long-run trend:**
  - Real GDP per capita grows as productivity rises.
- **Short-run fluctuations:**
  - Business cycles around the trend.
  - Recessions associated with:
    - lower output,
    - lower inflation,
    - higher unemployment.
- **Financial system:**
  - Channels saving into investment, supporting long-run growth.
  - Government borrowing can crowd out some private investment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Summary

- Long-run economic growth is driven by **labour productivity**, which depends on:
  - capital per worker,
  - human capital,
  - technological change.
- Potential GDP and the **output gap** help us see whether resources are over- or under-utilized.
- The **financial system** provides:
  - risk-sharing,
  - liquidity,
  - information.
- In a closed economy, **saving equals investment**:
  - budget deficits can crowd out private investment.
- The **business cycle** describes short-run fluctuations in real GDP, with:
  - expansions, peaks, recessions, and troughs,
  - characteristic movements in inflation and unemployment.

Thank you!